

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026
Comprehensive Examination

EC-3 Makeup Examination

Course No. : DSECL ZC416
Course Title : Mathematical Foundations for Data Science
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 hrs 30 mts
Date of Exam : 08-03-2026

No. of Pages = 2

No of questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Notation used here are as per the text book.

Q1. Consider the matrix $A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & -1 & 0 & 5 \end{bmatrix}$

(a) Find all the eigen values of A. Find the eigen space V corresponding to the largest eigen value of A. Also find a basis and dimension of V. [1 + 2 + 2 = 5M]

Since the given matrix A is a lower triangular matrix, the eigen values are the diagonal elements: -2, 3, -1, 5 [1M]

The largest eigen value is 5. So we shall find the eigenspace corresponding to $\lambda = 5$:

Solve $(A - 5I)X = 0$

$$A - 5I = \begin{bmatrix} -7 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 1 & -6 & 0 \\ -1 & -1 & 0 & 0 \end{bmatrix}$$

Let $X = (x_1, x_2, x_3, x_4)^T$.

Now from $(A - 5I)X = 0$ we have

1. $-7x_1 = 0 \Rightarrow x_1 = 0$
2. $-x_1 - 2x_2 = 0 \Rightarrow x_2 = 0$
3. $x_2 - 6x_3 = 0 \Rightarrow x_3 = 0$
4. $-x_1 - x_2 = 0$ [1M]

Since 4th column does not have a pivot element, x_4 is the free variable.

Hence any solution can be written as $X = \begin{bmatrix} 0 \\ 0 \\ 0 \\ r \end{bmatrix}$ where r is a real number.

Hence the eigen space corresponding to the eigen value 5 = $\text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right\}$. [1M]

Hence Basis = $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ and dimension = 1. [1M]

(b) Are the columns of A linearly independent. Justify your answer. [2M]

The column vectors are

$$v_1 = \begin{bmatrix} -2 \\ -1 \\ 0 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 3 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \end{bmatrix}$$

Vectors are **linearly independent** if

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0 \text{ implies } a_1 = a_2 = a_3 = a_4 = 0$$

$$a_1 \begin{bmatrix} -2 \\ -1 \\ 0 \\ -1 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 3 \\ 1 \\ -1 \end{bmatrix} + a_3 \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} + a_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad [1M]$$

This gives

$$\begin{bmatrix} -2a_1 \\ -a_1 + 3a_2 \\ a_2 - a_3 \\ -a_1 - a_2 + 5a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2a_1 = 0 \Rightarrow a_1 = 0$$

$$-a_1 + 3a_2 = 0 \quad 3a_2 = 0 \Rightarrow a_2 = 0$$

$$a_2 - a_3 = 0 \text{ implies } a_3 = 0$$

$$-a_1 - a_2 + 5a_4 = 0 \text{ implies } 5a_4 = 0 \Rightarrow a_4 = 0$$

Therefore, the columns of A are linearly independent. [1M]

Note: If any body computes $\det(A) = 30 \neq 0$, and conclude the columns are linearly independent give full 2 marks.

Q2. (a) Prove that the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is a symmetric positive definite matrix. [2M]

It is obvious that $A^T = A$ hence it is symmetric. [0.5M]

First principal minor: $D_1 = | 3 | = 3 > 0$

Second principal minor $D_2 = \det \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} = 3 - 1 = 2 > 0$. [0.5M]

Third principal minor $D_3 = \det \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2 \det \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} = 2 * 2 = 4 > 0$
 $= 2(2) = 4 > 0$

Since all principal minors are positive, A is positive definite. [1M]

Note: If any one proves that $X^T A X > 0$ for each $X \in \mathbb{R}^3$, award full 2 marks.

(b) Using this matrix define an inner product on $\mathbb{R}^3$. [1M]

For any vectors $X = (x_1, x_2, x_3)^T, Y = (y_1, y_2, y_3)^T \in \mathbb{R}^3$

Define $\langle X, Y \rangle = X^T A Y$ [0.5M]

Compute explicitly:

First compute Ay : $Ay = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3y_1 + y_2 \\ y_1 + y_2 \\ 2y_3 \end{bmatrix}$

Now $\langle X, Y \rangle = [x_1 \ x_2 \ x_3] \begin{bmatrix} 3y_1 + y_2 \\ y_1 + y_2 \\ 2y_3 \end{bmatrix} = x_1(3y_1 + y_2) + x_2(y_1 + y_2) + 2x_3y_3$ [0.5M]

(c) Show that the vectors $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ are not orthogonal with respect the inner product you defined on $\mathbb{R}^3$. [2M]

The given vectors are $u = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, v = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$

$Av = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

$$\begin{aligned}
\langle u, v \rangle &= u^T A v \\
&= [1 \ 2 \ -1] \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} [1M] \\
&= 1(5) + 2(1) + (-1)(0) = 7 \neq 0
\end{aligned}$$

Hence the given vectors are not orthogonal. [1M]

(d) Using the Gram–Schmidt orthonormalization process, construct an orthonormal set from the vectors v_1 and v_2 with respect to the inner product defined in part (b) [3M]

$$\text{The given vectors are } v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}.$$

Take $v_1 = v_1$

Compute $\langle v_1, v_1 \rangle = v_1^T A v_1$

$$\text{First } A v_1 = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$$

$$\text{Then } \langle v_1, v_1 \rangle = [1 \ 2 \ -1] \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = 5 + 6 + 2 = 13$$

$$\text{Normalize: } e_1 = \frac{u_1}{\sqrt{13}} = \frac{1}{\sqrt{13}} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} [1M]$$

Gram–Schmidt formula: $u_2 = v_2 - \frac{\langle v_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$

$$\text{Compute } \langle v_2, v_1 \rangle = [2 \ -1 \ 0] \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = 10 - 3 = 7$$

$$\text{Thus } u_2 = v_2 - \frac{7}{13} v_1 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} - \frac{7}{13} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{19}{13} \\ -\frac{27}{13} \\ \frac{7}{13} \end{bmatrix} [1M]$$

Normalizing u_2 :

$$A u_2 = \begin{bmatrix} \frac{30}{13} \\ -\frac{8}{13} \\ \frac{14}{13} \end{bmatrix}$$

$$\langle u_2, u_2 \rangle = \begin{bmatrix} \frac{19}{13} & -\frac{27}{13} & \frac{7}{13} \end{bmatrix} \begin{bmatrix} \frac{30}{13} \\ -\frac{8}{13} \\ \frac{14}{13} \end{bmatrix} = \frac{884}{169}$$

$$\text{Thus } \|u_2\| = \sqrt{\frac{884}{169}} = \frac{\sqrt{884}}{13}$$

$$\text{Normalize } e_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{884}} \begin{bmatrix} 19 \\ -27 \\ 7 \end{bmatrix}$$

Final Orthonormal Set

$$\boxed{e_1 = \frac{1}{\sqrt{13}} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, e_2 = \frac{1}{\sqrt{884}} \begin{bmatrix} 19 \\ -27 \\ 7 \end{bmatrix}} \quad [1M]$$

3. Consider the loss function $L(w_1, w_2) = w_1^4 + w_2^4 - (w_1 + w_2)$. Suppose we want to minimize this function using the Gradient Descent algorithm with initial point $(w_1^0, w_2^0) = (0, 0)$. Define the line search function $\phi(\alpha) = L((w_1^0, w_2^0) + \alpha d)$ where $d = -\nabla L(0, 0)$.

(a) Apply the binary search method on the interval $[0, 2]$ to determine the learning rate. Compute the reduced interval after two iterations. [4M]

Given the loss function $L(w_1, w_2) = w_1^4 + w_2^4 - (w_1 + w_2)$

Initial point $(w_1^0, w_2^0) = (0, 0)$

The gradient of L , $\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \begin{bmatrix} 4w_1^3 - 1 \\ 4w_2^3 - 1 \end{bmatrix}$

At $(0, 0)$, $\nabla L(0, 0) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

Thus, the descent direction $d = -\nabla L(0, 0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

The line search function $\phi(\alpha) = L((w_1^0, w_2^0) + \alpha d)$

$$\begin{aligned} &= L(\alpha, \alpha) = \alpha^4 + \alpha^4 - (\alpha + \alpha) \\ &= 2\alpha^4 - 2\alpha \end{aligned}$$

(a) Binary search on $[0, 2]$

First compute

$$\phi'(\alpha) = 8\alpha^3 - 2$$

Iteration 1

Midpoint $\alpha_1 = \frac{0+2}{2} = 1$

$\phi'(1) = 8(1) - 2 = 6 > 0$. Hence ϕ is increasing at 1.

[Some students might have compared $\phi(1)$ and $\phi(1.001)$ [some nearest point] and conclude ϕ is increasing at 1. Please award full mark]

Hence the minimum lies in $[0, 1]$. [2M]

Iteration 2

Midpoint $\alpha_2 = \frac{0+1}{2} = 0.5$

$\phi'(0.5) = 8(0.125) - 2 = 1 - 2 = -1 < 0$. Hence ϕ is decreasing at 0.5

Minimum lies New interval: $[0.5, 1]$. [2M]

(b) Using the second derivative test, determine the minimum of $\phi(\alpha)$ and verify whether the reduced interval obtained in part (a) is shrinking toward the optimal value. [3M]

Set $\phi'(\alpha) = 8\alpha^3 - 2 = 0$ implies $\alpha = \left(\frac{1}{4}\right)^{1/3} = 0.63$. [1M]

Second derivative: $\phi''(\alpha) = 24\alpha^2 = 24 * (0.63)^2 > 0$.

Hence $\alpha = 0.63$ is the point of minimum. [1.5M]

Verification: $0.63 \in [0.5, 1]$

Thus the binary search interval is shrinking toward the optimal learning rate. [0.5M]

Q4. Consider the function $J(w_0, w_1) = (w_0 - 3)^2 + (w_1 - 3)^2$.

We wish to minimize $J(w_0, w_1)$ using the Gradient Descent algorithm with momentum. The initial parameter values are $(w_0^0, w_1^0) = (2, 2)$, the learning rate is $\eta = 0.05$, the momentum parameter is $\beta = 0.5$, and the initial velocity vector is $(v_0^0, v_1^0) = (0, 0)$. Compute the parameter value (w_0^1, w_1^1) after the first iteration and (w_0^2, w_1^2) after the second iteration. [4M]

We want to minimize the function $J(w_0, w_1) = (w_0 - 3)^2 + (w_1 - 3)^2$

Initial values: $(w_0^0, w_1^0) = (2, 2)$

Learning rate: $\eta = 0.05$

Momentum parameter: $\beta = 0.5$

Initial velocity: $(v_0^0, v_1^0) = (0, 0)$

$$\text{Gradient of } J = \nabla J = \begin{bmatrix} 2(w_0 - 3) \\ 2(w_1 - 3) \end{bmatrix}$$

First Iteration:

$$\nabla J(2,2) = \begin{bmatrix} 2(2 - 3) \\ 2(2 - 3) \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

Velocity update

$$v^1 = \beta v^0 - \eta \nabla J(w^0)$$

$$v^1 = 0.5(0,0) - 0.05(-2, -2) = (0.1, 0.1) \text{ [1M]}$$

Parameter update

$$w^1 = w^0 + v^1$$

$$w^1 = (2, 2) + (0.1, 0.1) = (2.1, 2.1) \text{ [1M]}$$

Second Iteration

$$\nabla J(2.1,2.1) = \begin{bmatrix} 2(2.1 - 3) \\ 2(2.1 - 3) \end{bmatrix} = \begin{bmatrix} -1.8 \\ -1.8 \end{bmatrix}$$

Velocity update

$$v^2 = \beta v^1 - \eta \nabla J(w^1)$$

$$\begin{aligned} v^2 &= 0.5(0.1, 0.1) - 0.05(-1.8, -1.8) \\ &= (0.05, 0.05) + (0.09, 0.09) = (0.14, 0.14) \text{ [1M]} \end{aligned}$$

Parameter update

$$w^2 = w^1 + v^2$$

$$= (2.1, 2.1) + (0.14, 0.14) = (2.24, 2.24) . \text{ [1M]}$$

Q5. Consider the dataset

$$X = \begin{bmatrix} 0 & 3 \\ 1 & 5 \\ 2 & 7 \end{bmatrix},$$

where each row represents an observation in $\mathbb{R}^2$ and the columns represent features.

1. Compute the covariance matrix S of the dataset X . [2M]

1. Covariance Matrix S

First compute the mean of each feature:

$$\bar{x}_1 = \frac{0+1+2}{3} = 1$$

$$\bar{x}_2 = \frac{3+5+7}{3} = 5$$

Centre the data: We subtract the mean from each feature value.

$$X_c = \begin{bmatrix} 0-1 & 3-5 \\ 1-1 & 5-5 \\ 2-1 & 7-5 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{aligned} \text{The Covariance matrix } S &= \frac{1}{3} X_c^T X_c = \frac{1}{3} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix} \quad [1M] \\ &= \begin{bmatrix} \frac{2}{3} & \frac{4}{3} \\ \frac{4}{3} & \frac{8}{3} \end{bmatrix} \quad [1M] \end{aligned}$$

Some might have used directly the covariance formula from statistics and computed the covariance matrix. Please award full mark.

2. Determine the eigenvalues and eigenvectors of S . [3M]

To find the eigen values we solve $|S - \lambda I| = 0$

$$\det \begin{bmatrix} \frac{2}{3} - \lambda & \frac{4}{3} \\ \frac{4}{3} & \frac{8}{3} - \lambda \end{bmatrix} = 0$$

$$\left(\frac{2}{3} - \lambda\right) \left(\frac{8}{3} - \lambda\right) - \frac{16}{9} = 0$$

$$\lambda^2 - \frac{10}{3}\lambda = 0$$

$$\lambda \left(\lambda - \frac{10}{3}\right) = 0$$

So the eigen values are 0, 10/3. [1M]

Eigenvector for $\lambda = \frac{10}{3}$:

$$\left(S - \frac{10}{3}I\right)v = 0$$

$$\begin{bmatrix} \frac{2}{3} - \frac{10}{3} & \frac{4}{3} \\ \frac{4}{3} & \frac{8}{3} - \frac{10}{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -\frac{8}{3} & \frac{4}{3} \\ \frac{4}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $y = 2x$

Eigenvectors are $\left\{ r \begin{bmatrix} 1 \\ 2 \end{bmatrix} : r \neq 0 \right\}$ [1M] Note: If the answer is not given in the final form reduce 0.5M.

Eigenvector for $\lambda = 0$

$$\begin{bmatrix} \frac{2}{3} & \frac{4}{3} \\ \frac{4}{3} & \frac{8}{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $x + 2y = 0$ ie $x = -2y$

Eigenvectors are $\left\{ r \begin{bmatrix} -2 \\ 1 \end{bmatrix} : r \neq 0 \right\}$ [1M] Note: If the answer is not given in the final form reduce 0.5M.

3. Using Principal Component Analysis (PCA), identify the one-dimensional principal subspace of $\mathbb{R}^2$ onto which the data should be projected. [2M]

In **PCA**, the principal component corresponds to the **largest eigenvalue**.

Largest eigenvalue = $\frac{10}{3}$ [1M]

One-dimensional principal subspace span $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$. [1M]

Q6. Consider the following training dataset in $\mathbb{R}^2$:

Point	x_1	x_2	Class y
X_1	3	3	+1
X_2	5	5	+1
X_3	3	0	-1
X_4	5	0	-1

(a) Let $w = (w_1, w_2)^T \in \mathbb{R}^2$ be the weight vector, $b \in \mathbb{R}$ be the bias term, $\xi_i \geq 0, i = 1, 2, 3, 4$, be the slack variables, and $C > 0$ be the regularization parameter. Write the primal optimization problem for the soft-margin Support Vector Machine for the given dataset. [1M]

The soft-margin SVM primal problem is

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^4 \xi_i$$

subject to

$$y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, 2, 3, 4, \quad \xi_i \geq 0 \quad [0.5M]$$

Elaborating this we have

$$\min_{w, b, \xi} \frac{1}{2} (w_1^2 + w_2^2) + C(\xi_1 + \xi_2 + \xi_3 + \xi_4)$$

subject to

$$\begin{aligned}
3w_1 + 3w_2 + b &\geq 1 - \xi_1 \\
5w_1 + 5w_2 + b &\geq 1 - \xi_2 \\
-(3w_1 + b) &\geq 1 - \xi_3 \\
-(5w_1 + b) &\geq 1 - \xi_4 \\
\xi_i &\geq 0, i = 1, 2, 3, 4
\end{aligned}
\quad [0.5M]$$

(b) Show that the soft-margin SVM problem can be written in the unconstrained form:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^4 \max(0, 1 - y_i(w^T x_i + b)). \quad [2M]$$

From the constraint we have $y_i(w^T x_i + b) \geq 1 - \xi_i$ $i = 1, 2, 3, 4$

$$\text{rearrange: } \xi_i \geq 1 - y_i(w^T x_i + b)$$

But slack variables must also satisfy $\xi_i \geq 0$ [1M]

Thus ξ_i must be **at least the larger of the two quantities**: $0, 1 - y_i(w^T x_i + b)$

Therefore

$\xi_i = \max(0, 1 - y_i(w^T x_i + b))$ which is called the hinge loss.

Hence the soft-margin SVM problem can be written in the unconstrained form:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^4 \max(0, 1 - y_i(w^T x_i + b)). \quad [1M]$$

(c) Suppose $w = (1, -1)$ and $b = 0$.

1. Compute $y_i(w^T x_i + b)$ for each data point. [2M]

$$y_1(w^T \begin{bmatrix} 3 \\ 3 \end{bmatrix} + b) = 1[1 \quad -1] \begin{bmatrix} 3 \\ 3 \end{bmatrix} + 0 = 0$$

$$y_2(w^T \begin{bmatrix} 5 \\ 5 \end{bmatrix} + b) = 1[1 \quad -1] \begin{bmatrix} 5 \\ 5 \end{bmatrix} + 0 = 0$$

$$y_3(w^T \begin{bmatrix} 3 \\ 0 \end{bmatrix} + b) = -1[1 \quad -1] \begin{bmatrix} 3 \\ 0 \end{bmatrix} + 0 = -3$$

$$y_4(w^T \begin{bmatrix} 5 \\ 0 \end{bmatrix} + b) = -1[1 \quad -1] \begin{bmatrix} 5 \\ 0 \end{bmatrix} + 0 = -5 \quad [4 \times 0.5 = 2M]$$

2. Compute the hinge loss for each point. [1M]

$$\max(0, 1 - y_1(w^T \begin{bmatrix} 3 \\ 3 \end{bmatrix} + b)) = \max(0, 1 - 0) = 1$$

$$\max(0, 1 - y_2(w^T \begin{bmatrix} 5 \\ 5 \end{bmatrix} + b)) = \max(0, 1 - 0) = 1 \quad [0.5M]$$

$$\max(0, 1 - y_3(w^T \begin{bmatrix} 3 \\ 0 \end{bmatrix} + b)) = \max(0, 1 - (-3)) = 4$$

$$\max(0, 1 - 4(w^T \begin{bmatrix} 5 \\ 0 \end{bmatrix} + b)) = \max(0, 1 - (-5)) = 6 \quad [0.5M]$$

3. Determine the total objective value for $C = 1$. [1M]

$$\begin{aligned} \text{The total objective value} &= \frac{1}{2} \|w\|^2 + 1 \sum_{i=1}^4 \max(0, 1 - y_i(w^T x_i + b)) \\ &= \frac{1}{2} [1 + 1] + 1 + 1 + 4 + 6 = 13. \quad [1M] \end{aligned}$$
